

# Nikko J. Cleri

Eberly Postdoctoral Fellow  
The Pennsylvania State University

## Summary

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|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research:      | Galaxy Evolution, High-Redshift Galaxies, Emission-Line Galaxies, Interstellar Medium, Active Galactic Nuclei, Black Hole Seeds, Population III Stars, Star Formation, Dust Attenuation |
| Techniques:    | UV/Optical/Near-IR Spectroscopy, Photoionization Modeling                                                                                                                               |
| Proposals:     | As PI: 2 JWST, 1 HST, 29.2 JWST hours and >0.5M USD awarded<br>As Co-I: 14 JWST, 1 Gemini, 1 HET, >850 total hours awarded                                                              |
| Publications:  | 8 as lead/co-lead author, 101 total, 7559 citations, 43 h-index                                                                                                                         |
| Presentations: | 23 research, 17 outreach and professional development                                                                                                                                   |

## Academic and Professional Appointments

|         |                            |            |
|---------|----------------------------|------------|
| 2024-   | Eberly Postdoctoral Fellow | Penn State |
| 2021-24 | Graduate Student (Ph.D.)   | Texas A&M  |
| 2019-21 | Graduate Student (M.S.)    | UConn      |
| 2017-20 | Research Assistant (B.S.)  | UConn      |
| 2018    | NSF REU Student            | Texas A&M  |

## Education

|                                                                                                                     |                           |             |
|---------------------------------------------------------------------------------------------------------------------|---------------------------|-------------|
| <b>Ph.D. Astronomy</b>                                                                                              | Texas A&M University      | 2021 - 2024 |
| Advisor: Casey Papovich                                                                                             |                           |             |
| Thesis: <i>Spectroscopic Studies of Stars and Black Holes Across Cosmic Time</i>                                    |                           |             |
| <b>M.S. Physics</b>                                                                                                 | University of Connecticut | 2019 - 2021 |
| Advisor: Jonathan R. Trump                                                                                          |                           |             |
| Thesis: <i>CLEAR: Paschen-<math>\beta</math> Star Formation Rates and Dust Attenuation in Low Redshift Galaxies</i> |                           |             |
| <b>B.S. Physics   Mathematics Minor</b>                                                                             | University of Connecticut | 2015 - 2019 |
| Advisor: Gerald V. Dunne                                                                                            |                           |             |
| Undergraduate Research: <i>Resurgent trans-series for generalized Hastings-McLeod solutions</i>                     |                           |             |

## Awarded Proposals and Grants

|                        |                                                                                                               |                   |
|------------------------|---------------------------------------------------------------------------------------------------------------|-------------------|
| Principal Investigator |                                                                                                               | 3                 |
| 2026                   | JWST-GO-10518: <i>CEDAR: Characterizing Electron Densities At Reionization</i>                                | 29.2 hours/\$234k |
| 2024                   | JWST-AR-5558: <i>A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time</i>                   | \$174k            |
| 2021                   | HST-AR-16609: <i>Peering Through the Dust: Paschen-beta Indicators of Star Formation and Dust Attenuation</i> | \$136k            |

| Co-Investigator |                                                                                                                                                                                                       | 16           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 2026            | JWST-GO-11376: <i>Water in Little Red Dots? A Pathfinder Program with the MIRI/LRS to Confirm the Cool Temperatures of LRDs</i> (PI: B. Wang, J. Helton)                                              | 13.0 hours   |
| 2026            | JWST-GO-10445: <i>Reddened by dust or gas? Unveiling the "Red" in Little Red Dots with MIRI/LRS</i> (PI: A. de Graaff, D. Setton)                                                                     | 49.1 hours   |
| 2026            | JWST-GO-10311: <i>Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU</i> (PI: P. Rinaldi, E. Iani)                                                           | 26.7 hours   |
| 2025            | HET-PSU26-1-021: <i>The Hobby-Eberly Telescope Void Galaxy Sample</i> (PI: O. Curtis)                                                                                                                 | 26 hours     |
| 2025            | JWST-GO-8047: <i>Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features</i> (PI: B. Wang and E. Nelson) | 19.6 hours   |
| 2025            | JWST-GO-7488: <i>Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at <math>z=7.3</math></i> (PI: A. Weibel)                                                              | 14.1 hours   |
| 2025            | JWST-GO-8559: <i>SPAM: Star-formation from Photometry through the Addition of Medium-bands</i> (PI: K. Davis and R. Larson)                                                                           | 62.8 hours   |
| 2025            | JWST-GO-8410: <i>A Census of Galaxy Kinematics and Outflows to <math>z \sim 7</math></i> (PI: R. Simons)                                                                                              | 110.0 hours  |
| 2025            | JWST-GO-8204: <i>Give me a break: the search for stars in a prototypical Little Red Dot</i> (PI: J. Greene and I. Labbe)                                                                              | 17.4 hours   |
| 2024            | JWST-GO-5718: <i>A Spectroscopic Census of Faint, Broad-Line AGN at <math>z&gt;5</math></i> (PI: D. Kocevski and J. Guo)                                                                              | 20.49 hours  |
| 2024            | JWST-GO-5943: <i>What really are the Physical Properties of Galaxies in the Epoch of Reionization?</i> (PI: C. Papovich and W. Hu, T. Hutchison)                                                      | 61.83 hours  |
| 2024            | JWST-GO-5407: <i>MEOW: The MIRI Early Obscured-AGN Wide Survey</i> (PI: G. Leung and R. Endsley, S. Finkelstein)                                                                                      | 73.95 hours  |
| 2024            | JWST-GO-5507: <i>Deep Spectroscopy of Galaxies at <math>z=4-14</math>: Uncovering Drivers of Early Galaxy Formation and Black Hole Growth</i> (PI: T. Hutchison and R. Larson)                        | 23.29 hours  |
| 2024            | JWST-GO-6368: <i>The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS)</i> (PI: M. Dickinson)                                                                                                  | 293.21 hours |
| 2023            | JWST-GO-3703: <i>Breaking the <math>z=10</math> barrier with MIRI: redshift confirmation and detection of rest-frame optical emission lines</i> (PI: J. Zavala)                                       | 24.33 hours  |
| 2023            | GS-2023A-Q-136: <i>Optical Spectroscopy of JWST ERO Galaxies</i> (PI: B. Backhaus)                                                                                                                    | 20 hours     |

## Honors and Awards

|         |                                                         |            |
|---------|---------------------------------------------------------|------------|
| 2024    | Dean's Climate and Diversity Award (Group)              | Penn State |
| 2022    | Texas Space Grant Consortium Graduate Fellowship - \$5K | Texas A&M  |
| 2018    | NSF REU - \$5K                                          | Texas A&M  |
| 2016    | Dean's List - College of Liberal Arts and Sciences      | UConn      |
| 2015-19 | Governor's Scholarship - \$8.5K/yr                      | UConn      |
| 2015    | Community Service Scholarship - \$1K                    | UConn      |

## Teaching Experience

### Instructor of Record

|                                            |            |
|--------------------------------------------|------------|
| ASTRO 6: Stars, Galaxies, and the Universe | Penn State |
|--------------------------------------------|------------|

### Guest Lecturer

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|-------------------------------------|------------|
| ASTRO 502: Fundamental Astrophysics | Penn State |
|-------------------------------------|------------|

### Teaching Assistant/Course Assistant

|                                      |       |
|--------------------------------------|-------|
| PHYS 1501: Physics for Engineers I   | UConn |
| PHYS 1025: Introduction to Astronomy | UConn |

## Service

### Physics & Astronomy Community

Organizing Committee - 8th IGC Neighborhood Workshop, State College, PA, April 2026

Referee – Astronomy & Astrophysics (A&A)

Referee – Astrophysical Journal (ApJ)

### College

|                                                                           |            |
|---------------------------------------------------------------------------|------------|
| Astronomy Representative – Eberly College of Science Postdoctoral Council | Penn State |
|---------------------------------------------------------------------------|------------|

### Department

|                                          |            |
|------------------------------------------|------------|
| Founder/Organizer – ExGal Journal Club   | Penn State |
| Member – Climate and Diversity Committee | Penn State |
| Graduate Representative                  | Texas A&M  |
| Organizer – Astronomy Journal Club       | Texas A&M  |

## Mentoring

### Students Supervised

|                                                   |       |
|---------------------------------------------------|-------|
| Primary mentor for 3 undergraduates at Penn State | 2024- |
|---------------------------------------------------|-------|

### Other Mentoring Activities

|                                                                                          |            |
|------------------------------------------------------------------------------------------|------------|
| Co-Founder – Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN) |            |
| Primary Mentor – Penn State-Nanjing Exchange Program                                     | Penn State |
| Coordinator – Mentoring and Advising Graduates in an Inclusive Community (MAGIC)         | Texas A&M  |
| Mentor – Mentoring and Advising Graduates in an Inclusive Community (MAGIC)              | Texas A&M  |
| Mentor – UConn Undergraduate Peer Mentoring                                              | UConn      |

## Outreach

|                                            |            |
|--------------------------------------------|------------|
| Presenter – Astronomy on Tap State College | Penn State |
| Volunteer – Gateway to Graduate School     | Texas A&M  |

|                                                 |           |
|-------------------------------------------------|-----------|
| Demonstrator – Physics and Engineering Festival | Texas A&M |
| High School Research Reviewer – Lumiere         | Texas A&M |
| Presenter – Astronomy on Tap BCS ‘In the News’  | Texas A&M |
| Treasurer – Astronomy on Tap BCS                | Texas A&M |
| Pen-Pal – Letters to a Pre-Scientist            | Texas A&M |
| Volunteer – Mitchell Institute Star Party Group | Texas A&M |

## Press

A Week Downeast, Challenging Assumptions about Galaxy Evolution, Astrobites, O. Cooper et al. 2025

## Collaborations and Survey Membership

### Co-Investigator

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|---------------------------------------------------------------------------|------|
| SPAM: Star-formation from Photometry through the Addition of Medium-bands | JWST |
| CAPERS: The CANDELS-Area Prism Epoch of Reionization Survey               | JWST |
| THRILS: The High [Redshift+Ionization] Line Search                        | JWST |

### Contributing Member

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|-----------------------------------------------------------------------------|------|
| LEGGOS: Lensing and Galaxy Growth: Observing Substructures                  | JWST |
| POPPIES: The Public Observation Pure Parallel Infrared Emission-Line Survey | JWST |
| RUBIES: Red Unknowns: Bright Infrared Extragalactic Survey                  | JWST |
| NGDEEP: The Next Generation Deep Exploratory Public Survey                  | JWST |
| CEERS: The Cosmic Evolution Early Release Science Survey                    | JWST |
| CLEAR: The CANDELS Ly $\alpha$ Emission at Reionization Survey (inactive)   | HST  |

## Technical Skills and Programming Languages

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|-----------|----------------------------------------------------------------------------|
| Languages | Fluent: Python, LaTeX                                                      |
|           | Familiar: SQL, Julia, C, C++, R, IDL, perl, Mathematica, MATLAB, HTML, CSS |
| Software  | Fluent: Cloudy, PyNeb, LiMe                                                |
|           | Familiar: grizli, DS9, IRAF, sbatch, slurm                                 |

## Website Architect

Personal Website: njcleri.github.io  
Mentoring and Advising Graduates in an Inclusive Community (MAGIC) (co-author): tx.ag/tamumagic  
Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN): penguin-mentoring.github.io

## Publications

A current list of my papers can be found on SciX here.

### Lead/Co-Lead/Student-Led Papers

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† = Undergraduate-led papers

8. **Cleri, N. J.**, et al. 2026, *RUBIES: The Evolution of the Ionization Parameter from  $0 < z < 9$* , arXiv e-prints, arXiv:2511.08671 (submitted to ApJL)
7. †Ge, S.-R., **Cleri, N. J.**, et al. 2025, *JWST/NIRSpec Reveals a Small Population of Dominant Dust-Obscured Ionizing Sources in Galaxies at  $1 < z < 3$* , arXiv e-prints, arXiv:2511.08671 (submitted to ApJ)
6. **Cleri, N. J.**, Olivier, G. M., Backhaus, B. E., et al. 2025, *Optical Strong Line Ratios Cannot Distinguish Between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities*, ApJ, 994, 146,
5. Backhaus, B. E., **Cleri, N. J.**, et al. 2025, *Emission-Line Diagnostics at  $z > 4$ :  $[O III]\lambda 4364/H\gamma$  versus  $[Ne III]\lambda 3870/[O II]\lambda 3730$* , ApJ, 994, 125
4. **Cleri, N. J.**, Olivier, G. M., Hutchison T. A., et al. 2023, *Using  $[Ne V]/[Ne III]$  to Understand the Nature of Extreme-Ionization Galaxies*, ApJ, 953, 10
3. **Cleri, N. J.**, Yang, G., Papovich, C., et al. 2023, *CLEAR: High-Ionization  $[Ne V] \lambda 3426$  Emission-line Galaxies at  $1.4 < z < 2.3$* , ApJ, 948, 112
2. **Cleri, N. J.**, Trump, J. R., Backhaus, B. E., et al. 2022, *CLEAR: Paschen- $\beta$  Star Formation Rates and Dust Attenuation of Low Redshift Galaxies*, ApJ, 929, 3
1. **Cleri, N. J.**, Dunne, G. V., 2020, *Resurgent trans-series for generalized Hastings-McLeod solutions*, Journal of Physics A: Mathematical General, 53, 355203

### Significant Author

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4. Larson, R. L., Finkelstein, S. L., Kocevski, D. D., Hutchison, T. A., Trump, J. R., Arrabal Haro, P., Bromm, V., **Cleri, N. J.**, et al. 2023, *A CEERS Discovery of an Accreting Supermassive Black Hole 570 Myr after the Big Bang: Identifying a Progenitor of Massive  $z > 6$  Quasars*, ApJL, 953, L29
3. Backhaus, B. E., Bridge J. S., Trump, J. R., **Cleri, N. J.**, et al. 2023, *CLEAR: Detecting Low-Luminosity Active Galactic Nuclei at  $0.6 < z < 1.3$  via Spatially Resolved Hubble Space Telescope Grism Emission Line Ratios*, ApJ, 943, 37
2. Prescott, M. K. M., Finlator, K. M., **Cleri, N. J.**, et al. 2022, *Using Multiple Emission Line Ratios to Constrain the Slope of the Dust Attenuation Law*, ApJ, 928, 71
1. Backhaus, B. E., Trump, J. R., **Cleri, N. J.**, et al. 2022, *CLEAR: Emission Line Ratios at Cosmic High Noon*, ApJ, 926, 161

### Co-Author: Published

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71. Papovich, C., et al. 2026, *Galaxies in the Epoch of Reionization Are All Bark and No Bite – Plenty of Ionizing Photons, Low Escape Fractions*, ApJ, 1000, 111
70. Hviding, R. E., et al. 2026, *The X-Ray Dot: Exotic Dust or a Late-Stage Little Red Dot?*, ApJL, 1000, L18
69. Gandolfi, G., et al. 2026, *Mysteries of Capotauro - investigating the puzzling nature of an extreme F356W-dropout*, A&A, 706, A364,4
68. Zhang, Y., et al. 2026, *RUBIES Spectroscopically Confirms the High Number Density of Quiescent Galaxies from  $2 < z < 5$* , ApJ, 997, 252

67. Setton, D. J., et al. 2025, *Little Red Dots at an Inflection Point: Ubiquitous “V-Shaped” Turnover Consistently Occurs at the Balmer Limit*, ApJ, 995, 118
66. Hviding, R. E., et al. 2025, *RUBIES: A Spectroscopic Census of Little Red Dots; All V-Shaped Point Sources Have Broad Lines*, A&A, 702, A57
65. Mascia, S., et al. 2025, *Little impact of mergers and galaxy morphology on the production and escape of ionizing photons in the early Universe*, A&A, 701, A122,
64. Leung, G. C. K., et al. 2024, *Exploring the Nature of Little Red Dots: Constraints on AGN and Stellar Contributions from PRIMER MIRI Imaging*, ApJ, 992, 26
63. Setton, D. J., et al. 2025, *A Confirmed Deficit of Hot and Cold Dust Emission in the Most Luminous Little Red Dots*, ApJL, 991, L10
62. Katz, H., et al. 2025, *21 Balmer Jump Street: The Nebular Continuum at High Redshift and Implications for the Bright Galaxy Problem, UV Continuum Slopes, and Early Stellar Populations*, The Open Journal of Astrophysics, 8, 104
61. Kokorev, V., et al. 2025, *CAPERS Observations of Two UV-Bright Galaxies at  $z > 10$ . More Evidence for Bursting Star Formation in the Early Universe*, ApJL, 988, L10
60. Burgarella, D., et al. 2025, *CEERS: Possibly forging the first dust grains in the universe: A population of galaxies with spectroscopically derived extremely low dust attenuation (GELDA) at  $4.0 < z < 11.4$* , A&A, 699, A336
59. Taylor, A. J., et al. 2025, *Broad-line AGNs at  $3.5 < z < 6$ : The Black Hole Mass Function and a Connection with Little Red Dots*, ApJ, 986, 165
58. Llerena, M. et al. 2025, *The ionizing photon production efficiency of star-forming galaxies at  $z \sim 4 - 10$* , A&A, 698, A302
57. Brooks, M., et al. 2024, *Here There Be (Dusty) Monsters: High-redshift Active Galactic Nuclei Are Dustier than Their Hosts*, ApJ, 986, 177
56. Kocevski, D. D., et al. 2025, *The Rise of Faint, Red Active Galactic Nuclei at  $z > 4$ : A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields*, ApJ, 986, 126
55. Wang, B., et al. 2025, *RUBIES: JWST/NIRSpec Confirmation of an Infrared-luminous, Broad-line Little Red Dot with an Ionized Outflow*, ApJ, 984, 121
54. de Graaff, A., et al. 2025, *RUBIES: a complete census of the bright and red distant Universe with JWST/NIRSpec*, A&A, 697, A189,
53. Finkelstein, S. L., et al. 2025, *The Cosmic Evolution Early Release Science Survey (CEERS)*, ApJL, 983, L4
52. Weibel, A., et al. 2025, *RUBIES Reveals a Massive Quiescent Galaxy at  $z=7.3$* , ApJ, 983, 11
51. Cooper, O. R., et al. 2025, *RUBIES: JWST/NIRSpec resolves evolutionary phases of dusty star-forming galaxies at  $z \sim 2$* , ApJ, 982, 125
50. de Graaff, A., et al. 2025, *Efficient formation of a massive quiescent galaxy at redshift 4.9*, Nature Astronomy 9, 280
49. Shen, L., et al. 2025, *NGDEEP: The Star Formation and Ionization Properties of Galaxies at  $1.7 < z < 3.4$* , ApJL, 980, L45
48. Cole, J. W., et al. 2025, *CEERS: Increasing Scatter along the Star-forming Main Sequence Indicates Early Galaxies Form in Bursts*, ApJ, 979, 193
47. Zavala, J., et al. 2025, *A luminous and young galaxy at  $z = 12.33$  revealed by a JWST/MIRI detection of  $H\alpha$  and  $[O III]$* , Nature Astronomy, 9, 155
46. Cheng, Y., et al. 2025, *Unveiling the Dark Side of UV/Optical Bright Galaxies: Optically Thick Dust Absorption*, ApJ, 979, 71

45. Bisigello, L., et al. 2025, *Spectroscopic confirmation of a dust-obscured, metal-rich dwarf galaxy at  $z \sim 5$* , A&A, 693, L18
44. Rose, C., et al. 2024, *CEERS Key Paper. IX. Identifying Galaxy Mergers in CEERS NIRCам Images Using Random Forests and Convolutional Neural Networks*, ApJL, 976, L8
43. Llerena, M., et al. 2024, *Physical properties of extreme emission-line galaxies at  $z \sim 4 - 9$  from the JWST CEERS survey*, A&A, 691, A59
42. Gupta, A. R., et al. 2024, *Emission-Line Ratios and Ionization Conditions of CEERS Star-Forming Galaxies with JWST/NIRSpec*, Research Notes of the American Astronomical Society, 8, 266
41. Davis, K., et al. 2024, *A Census from JWST of Extreme Emission-line Galaxies Spanning the Epoch of Reionization in CEERS*, ApJ, 974, 42
40. Calabró, A., et al. 2024, *The evolution of the star formation rate and  $\Sigma_{SFR}$  of galaxies in cosmic morning ( $4 < z < 10$ )*, A&A, 690, A290
39. Chworowsky, K., et al. 2024, *Evidence for a Shallow Evolution in the Volume Densities of Massive Galaxies at  $z = 4$  to 8 from CEERS*, AJ, 168, 113
38. Seillé, L.-M., et al. 2024, *Physical properties of strong  $1 < z < 3$  Balmer and Paschen lines emitters observed with JWST*, A&A, 689, A102
37. Hu, W., et al. 2024, *Characterizing the Average Interstellar Medium Conditions of Galaxies at  $z \sim 5.6-9$  with UV and Optical Nebular Lines*, ApJ, 971, 21
36. Napolitano, L., et al. 2024, *'Peering into cosmic reionization: the  $Ly\alpha$  visibility evolution from galaxies at  $z = 4.5-8.5$  with JWST*, A&A, 688, A106
35. Ronayne, K., et al. 2024, *CEERS:  $7.7 \mu m$  PAH Star Formation Rate Calibration with JWST MIRI*, ApJ, 970, 61
34. Wang, B., et al. 2024, *RUBIES: Evolved Stellar Populations with Extended Formation Histories at  $z \sim 7 - 8$  in Candidate Massive Galaxies Identified with JWST/NIRSpec*, ApJL, 969, L13
33. Finkelstein, S. L., et al. 2024, *The Complete CEERS Early Universe Galaxy Sample: A Surprisingly Slow Evolution of the Space Density of Bright Galaxies at  $z \sim 8.5 - 14.5$* , ApJL, 969, L2
32. Pirzkal, N., et al. 2024, *The Next Generation Deep Extragalactic Exploratory Public Near-Infrared Slitless Survey Epoch 1 (NGDEEP-NISS1): Extra-Galactic Star-formation and Active Galactic Nuclei at  $0.5 < z < 3.6$* , ApJ, 969, 90
31. Jung, I., et al. 2024, *CEERS: Diversity of Lyman-Alpha Emitters during the Epoch of Reionization*, ApJ, 967, 73
30. Mascia, S. et al. 2024 *New insight on the nature of cosmic reionizers from the CEERS survey* A&A, 685, A3
29. Morales, A. M., et al. 2024, *Rest-Frame UV Colors for Faint Galaxies at  $z \sim 9 - 16$  with the JWST NGDEEP Survey*, ApJL, 964, L24
28. Cheng, Y., et al. 2024, *Exploring the Gas-Phase Metallicity Gradients of Star-forming Galaxies at Cosmic Noon*, ApJ, 964, 94
27. Shen, L., et al. 2024, *NGDEEP Epoch 1: Spatially Resolved  $H\alpha$  Observations of Disk and Bulge Growth in Star-Forming Galaxies at  $z \sim 0.6-2.2$  from JWST NIRISS Slitless Spectroscopy*, ApJL, 963, L49
26. Barro, G., et al. 2024, *Extremely Red Galaxies at  $z = 5-9$  with MIRI and NIRSpec: Dusty Galaxies or Obscured Active Galactic Nuclei?*, ApJ, 963, 128
25. Backhaus, B. E., et al. 2024, *CEERS Key Paper. VIII. Emission-line Ratios from NIRSpec and NIRCам Wide-Field Slitless Spectroscopy at  $z > 2$* , ApJ, 962, 195
24. Kirkpatrick, A., et al. 2023, *CEERS Key Paper VII: JWST/MIRI Reveals a Faint Population of Galaxies at Cosmic Noon Unseen by Spitzer*, ApJL, 959, L7

23. Calabró, A., et al. 2023, *Near-infrared emission line diagnostics for AGN from the local Universe to redshift 3*, A&A, 679, A80
22. Fujimoto, S., et al. 2023, *ALMA FIR View of Ultra High-redshift Galaxy Candidates at  $z \sim 11-17$ : Blue Monsters or Low- $z$  Red Interlopers?*, ApJ, 955, 130
21. Kocevski, D. D., et al. 2023, *Hidden Little Monsters: Spectroscopic Identification of Low-Mass, Broad-Line AGN at  $z > 5$  with CEERS*, ApJL, 954, L4
20. Arrabal Haro, P., et al. 2023, *Spectroscopic confirmation of CEERS NIRCам-selected galaxies at  $z \simeq 8 - 10$* , ApJL, 951, L22
19. Estrada-Carpenter, V., et al. 2023, *CLEAR: The Morphological Evolution of Galaxies in the Green Valley*, ApJ, 951, 115
18. Yang, G., et al. 2023, *CEERS Key Paper VI: JWST/MIRI Uncovers a Large Population of Obscured AGN at High Redshifts*, ApJL, 950, L5
17. Papovich, C., et al. 2023, *CEERS Key Paper IV: Galaxies at  $4 < z < 9$  are Bluer than They Appear – Characterizing Galaxy Stellar Populations from Rest-Frame  $\sim 1$  micron Imaging*, ApJL, 949, L18
16. Simons, R. C., et al. 2023, *CLEAR: Survey Overview, Data Analysis and Products*, ApJS, 266, 13
15. Constantin, L. et al. 2023, *Expectations of the size evolution of massive galaxies at  $3 \leq z \leq 6$  from the TNG50 simulation: the CEERS/JWST view*, ApJ, 946, 71
14. Perez-Gonzalez, P. G., et al. 2023, *CEERS Key Paper V: A triality on the nature of HST-dark galaxies*, ApJL, 946, L16
13. Kocevski, D. D., et al. 2023, *CEERS Key Paper II: The Resolved Host Properties of AGN at  $3 < z < 5$  with JWST*, ApJL, 946, L14
12. Finkelstein, S. L., et al. 2023, *CEERS Key Paper I: An Early Look into the First 500 Myr of Galaxy Formation with JWST*, ApJL, 946, L13
11. Guo, Y. et al. 2023, *First Look at  $z > 1$  Bars in the Rest-Frame Near-Infrared with JWST Early CEERS Imaging*, ApJL, 945, L10
10. Trump, J. R. et al. 2023, *The Physical Conditions of Emission-Line Galaxies at Cosmic Dawn from JWST/NIRSpec Spectroscopy in the SMACS 0723 Early Release Observations*, ApJ, 945, 35
9. García-Argumán, A. et al. 2023, *Probing the earliest phases in the formation of massive galaxies with simulated HST+JWST imaging data from Illustris*, ApJ, 944, 3
8. Zavala, J. et al. 2023, *Dusty starbursts masquerading as ultra high redshift galaxies in JWST observations*, ApJL, 943, L9
7. Rose, C. et al. 2023, *Identifying Galaxy Mergers in Simulated CEERS NIRCам Images using Random Forests*, ApJ, 942, 54
6. Finkelstein, S. L. et al. 2022, *A Long Time Ago in a Galaxy Far, Far Away: A Candidate  $z \sim 14$  Galaxy in Early JWST CEERS Imaging*, ApJL, 940, L55
5. Papovich, C. et al. 2022, *CLEAR: The Ionization and Chemical-Enrichment Properties of Galaxies at  $1.1 < z < 2.3$* , ApJ, 937, 22
4. Matharu, J. et al. 2022, *CLEAR: The Evolution of Spatially Resolved Star Formation in Galaxies between  $0.5 \leq z \leq 1.7$  using  $H\alpha$  Emission Line Maps*, ApJ, 937, 16
3. Jung, I. et al. 2022, *CLEAR: Boosted  $Ly\alpha$  Transmission of the Intergalactic Medium in UV bright Galaxies*, ApJ, 933, 87
2. Simons, R. C. et al. 2021, *CLEAR: The Gas-Phase Metallicity Gradients of Star-Forming Galaxies at  $0.6 < z < 2.6$* , ApJ, 923, 203



1. Estrada-Carpenter, V. et al. 2020, *CLEAR II: Evidence for Early Formation of the Most Compact Quiescent Galaxies at High Redshift*, ApJ, 880, 2

Co-Author: Submitted

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18. Hu W., et al. 2026, *Using Ly $\alpha$  Transmitted Spectrum to Probe IGM Transmission and Identify Ionized Structures in Cosmic Reionization*, arXiv e-prints, arXiv:2605.02539
17. Lambrides, E., et al. 2026, *The Glimmlr: Spectroscopic Variability in a  $z \sim 7$  LRD Indicates Rapid Changes in Both the Narrow and Broad Line Regions*, arXiv e-prints, arXiv:2604.25991
16. Davis, K., et al. 2026, *Extreme Emission Line Galaxies in CEERS Are Powered by Star Formation, not AGN*, arXiv e-prints, arXiv:2602.23310
15. Wang, B., et al. 2026, *Water absorption confirms cool atmospheres in two little red dots*, arXiv e-prints, arXiv:2602.06024
14. Arevalo-Gonzalez, F., et al. 2025, *The AGN nature of strong CIII emitters in the Early Universe with JWST*, arXiv e-prints, arXiv:2512.16365
13. Helton, J. M., et al. 2025, *Ionizing Photon Production Efficiencies and Chemical Abundances at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0*, arXiv e-prints, arXiv:2512.19695
12. Hutchison, T. A., et al. 2025, *THRILS – The High-(Redshift+Ionization) Line Search: Program Description & Redshift Catalog*, arXiv e-prints, arXiv:2512.12509
11. Lewis, Z., et al. 2025, *The Mass-Metallicity Relation and its Observational Effects at  $z \sim 3-6$* , arXiv e-prints, arXiv:2512.03134
10. Hutchison, T. A., et al. 2025, *Little Red Dots host Black Hole Stars: A unified family of gas-reddened AGN revealed by JWST/NIRSpec spectroscopy*, arXiv e-prints, arXiv:2512.02000
9. de Graaff, A., et al. 2025c, *Little Red Dots host Black Hole Stars: A unified family of gas-reddened AGN revealed by JWST/NIRSpec spectroscopy*, arXiv e-prints, arXiv:2511.21820
8. Brooks, M., et al. 2025b, *Beyond the Monsters: A More Complete Census of Black Hole Activity at Cosmic Dawn*, arXiv e-prints, arXiv:2511.19609
7. Llerena, M., et al. 2025, *Extreme equivalent width-selected low-mass starbursts at  $z = 4 - 9$ : insights into their role in cosmic reionization*, arXiv e-prints, arXiv:2510.25647
6. Perry, M. N., et al. 2025, *The Prevalence of Bursty Star Formation in Low-Mass Galaxies at  $z=1-7$  from H $\alpha$ -to-UV Diagnostics*, arXiv e-prints, arXiv:2510.05388
5. McConachie, I., et al. 2025, *Where Galaxies Go to Die: The Environments of Massive Quiescent Galaxies at  $3 < z < 5$* , arXiv e-prints, arXiv:2510.25024
4. Jones, B. L., et al. 2025, *The  $M_{\text{BH}} - M_*$  Relationship at  $3 < z < 7$ : Big Black Holes in Little Red Dots*, arXiv e-prints, arXiv:2510.07376
3. Lambrides, E., et al. 2025, *Discovery of Multiply Ionized Iron Emission Powered by an Active Galactic Nucleus in a  $z \sim 7$  Little Red Dot*, arXiv e-prints, arXiv:2509.09607
2. Wang, B., et al. 2025, *The Missing Hard Photons of Little Red Dots: Their Incident Ionizing Spectra Resemble Massive Stars*, arXiv e-prints, arXiv:2508.18358
1. Jung, I., et al. 2022, *New  $z > 7$  Lyman-alpha Emitters in EGS: Evidence of an Extended Ionized Structure at  $z \sim 7.7$* , arXiv e-prints, arXiv:2212.09850

## Presentations

Research Presentations (**bold** = invited)

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23. **Talk:** *The Rest-Frame Optical Spectra of High-Redshift Galaxies: What are we seeing?*, at the Dartmouth College, Hanover, New Hampshire, USA 1 May 2026
22. **Talk:** *The Rest-Frame Optical Spectra of High-Redshift Galaxies: What are we seeing?*, at the California Institute of Technology, virtual 4 May 2026
21. **Talk:** *Characterizing High-Redshift Sources with Optical Emission Line Ratios*, at the Space Telescope Science Institute, Baltimore, Maryland, USA 14 August 2025
20. **Talk:** *Characterizing High-Redshift Objects with Optical Strong Line Ratios*, Winter Harbor, Maine, USA 28 July 2025
19. **Talk:** *A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time*, Bergen, Netherlands 8 May 2025
18. **Talk:** *High-Redshift Diagnostics of Star Formation and Active Galactic Nuclei*, The Pennsylvania State University, State College, Pennsylvania, USA 10 February 2025
17. **Talk:** *High-Redshift Diagnostics of Star Formation and Active Galactic Nuclei*, The Pennsylvania State University, State College, Pennsylvania, USA 17 September 2024
16. **Talk:** *High-Redshift Diagnostics of Ionizing Sources*, University of Wisconsin, Madison, Wisconsin, USA 17 July 2024
15. **Discussion Chair:** *Active Galactic Nuclei and Little Red Dots*, San Lorenzo de El Escorial, Spain 13 May 2024
14. **Talk:** *Diagnostics of Ionizing Sources at High-z using Models and Observations*, San Lorenzo de El Escorial, Spain 13 May 2024
13. **Talk:** *Diagnostics of AGN, Black Hole Seeds, and Population III Stars with JWST* at the AAS 243rd Meeting, New Orleans, Louisiana, USA 10 January 2024
12. **Poster:** *Emission Line Ratio Diagnostics of AGN, Black Hole Seeds and Population III Stars with JWST* at the First Year of JWST Science Conference, Space Telescope Science Institute, Baltimore, Maryland, USA 11 September 2023
11. **Talk:** *Diagnostics of Exotic Ionizing Sources with JWST* at Texas A&M Astrosymposium, College Station, Texas, USA 17 August 2023
10. **Talk:** *Diagnostics of Exotic Ionizing Sources Across Cosmic Time - High-Ionization Emission-Line Ratios: Ne53* at University of Texas, Austin, Texas, USA 10 May 2023
9. **Poster:** *High-Ionization [Ne V] Emission-Line Galaxies at Cosmic Noon and the Epoch of Reionization* at AAS 241st Meeting, Seattle, Washington, USA 12 January 2023
8. **Talk:** *Using [Ne V] to Constrain the Sources of Highly-Energetic Photoionization Across Cosmic Time: Exploring the "Mystery of Neon" with HST and JWST* at Texas A&M University, College Station, Texas, USA 2 December 2022
7. **Talk:** *Extreme High-Ionization Emission-Line Galaxies at Cosmic Noon and the Epoch of Reionization: Exploring the "Mystery of Neon" with HST and JWST* at Texas A&M University, College Station, Texas, USA 18 August 2022
6. **Talk:** *The Evolution of Spectroscopy from HST to JWST: Implications for the Epoch of Reionization* at Texas A&M University, College Station, Texas, USA 22 July 2022

5. Poster: *HST Grism Observations of Paschen-Line Star-Formation and Dust Attenuation: A Precursor to the JWST Era* at AAS 240th Meeting, Pasadena, California, USA 14 June 2022
4. Talk: *Paschen- $\beta$  Star Formation Rates and Dust Attenuation with HST and JWST* at Texas A&M Astrosymposium, College Station, Texas, USA 27 August 2021
3. Poster: *CLEAR: Paschen- $\beta$  Star Formation Rates and Dust Attenuation in Low Redshift Galaxies* at AAS 237th Meeting, Virtual 13 January 2021
2. Poster: *Modeling  $^8B$  Solar Neutrino Detection with  $CE\nu NS$*  at AAS 233rd Meeting, Seattle, Washington, USA 9 January 2019
1. Poster: *Modeling  $^8B$  Solar Neutrino Detection with  $CE\nu NS$*  at TAMU Undergraduate Research Poster Session, College Station, Texas, USA 1 August 2018

#### Outreach Presentations

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7. "Broken Lecture": *Did JWST "Break The Universe"?*, State College, Pennsylvania, USA 29 April 2026
6. Talk: *Did JWST "Break The Universe"?*, State College, Pennsylvania, USA 11 April 2026
5. Talk: *Island Universes: Is the Milky Way Everything?* at Astronomy on Tap State College, State College, Pennsylvania, USA 19 March 2026
4. Talk: *Astronomy and You: The Impacts of Astronomy on Everyday Life* at Astronomy on Tap State College, State College, Pennsylvania, USA 20 March 2025
3. Talk: *The Evolving Universe Through JWST's Eyes* at the Pennsylvania State University, State College, Pennsylvania 1 February 2025
2. Talk: *The Origin of the Elements: Chemistry Across Cosmic Time* at Astronomy on Tap State College, State College, Pennsylvania, USA 19 September 2024
1. Talk: *Beyond the Telescope: Unraveling Mysteries with AI in Astronomy* at Astronomy on Tap B/CS, Bryan, Texas, USA 24 April 2024

#### Professional Development Presentations

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10. Panel: *Applying to Postdocs* at The Pennsylvania State University, State College, Pennsylvania, USA 7 April 2026
9. Panel: *Applying to Postdocs* at The Pennsylvania State University, State College, Pennsylvania, USA 24 March 2025
8. Panel: *GLASS Postdoc Panel* at Texas A&M University, College Station, Texas, USA 8 March 2024
7. Talk: *How to Be A Referee* at Texas A&M University, College Station, Texas, USA 10 November 2023
6. Panel: *How to Get Into Grad School* at Texas A&M University, College Station, Texas, USA 28 July 2023
5. Talk: *Data Visualization in Astronomy: More Important than the Science Itself?* at Texas A&M University, College Station, Texas, USA 11 November 2022
4. Panel: *How to Get Into Grad School* at Texas A&M University, College Station, Texas, USA 29 July 2022
3. *Data Visualization in Astronomy: More Important than the Science Itself?* at Texas A&M University, College Station, Texas, USA 2 June 2022

2. Workshop: *Matplotlib: The Champion of Plotting in Python* at Texas A&M University, College Station, Texas, USA 2 June 2022
1. Workshop: *pandas: Your Best Friend for Data Analysis in Python* at Texas A&M University, College Station, Texas, USA 1 June 2022

## References

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